



ENGINEERING MATHEMATICS-II

MATLAB

Department of Science and Humanities

Laplace Transforms

- In order to find the Laplace transform $F(s)$ of a function $f(t)$, we first need to specify the variables t and s using *$\text{syms } s \ t$*
- Define the function f
- Find the Laplace transform $f(t)$ by using $F = \text{laplace}(f, t, s)$
- To make the expression more readable one can use the commands, *$\text{simplify}(F)$*

Examples

Find the Laplace transforms of $f(t) = -1.25 + 3.5te^{-3t} + 1.25e^{-2t}$

```
>> syms t s
```

```
>> f= -1.25 + 3.5*t*exp(-3*t)+1.25*exp(-2*t);
```

```
>> F=laplace(f, t, s)
```

Output: $F = -5/(4*(s+2)) + 7/(2*(s+2)^2) - 5/(4*s)$

```
>> simplify(F)
```

Output: $\text{ans} = -(-2*s^2 + 16*s + 45)/(2*s*(s+2)*(s+3)^2)$

Examples

Find the Laplace transforms of $f(t) = e^{at}\sin(bt)$

```
>> syms t s a b
```

```
>> f= exp(a*t)*sin(b*t);
```

```
>> laplace(f, t, s)
```

Output: $b/(b^2+(a-s)^2)$

Examples

Find the Laplace transforms of $f(t) = ke^{-at}\cos(wt)$

```
>> syms t s k a w
```

```
>> T= laplace(k*exp(-a*t)*cos(w*t), t, s)
```

Output: $T = (k*(a + s))/((a + s)^2 + w^2)$

```
>> simplify(T)
```

ans = $(k*(a + s))/((a + s)^2 + w^2)$

Laplace transformation of Dirac Delta and Heaviside's Functions

Note that:

- Heaviside's Functions also called Unit step function
- It is denoted as $u(t - a)$ or $H(t - a)$

Using unit step function, find the Laplace transform of $f(t) = t - a$

```
>> syms t s
```

```
>> syms a positive
```

```
>> laplace heaviside(t-a),t,s)
```

Output: $\exp(-a*s)/s$

Laplace transformation of Dirac Delta and Heaviside's Functions

Using unit step function, find the Laplace transform of $f(t) = \sin 2t - \sin 3t$

```
>> syms t s;
```

```
>> laplace heaviside(t)*sin(2*t)-sin(3*t))
```

Output: $2/(s^2 + 4) - 3/(s^2 + 9)$

Dirac delta Functions

Note that:

- Dirac delta function also called Unit impulse function
- It is denoted as $\delta(t - a)$

Using unit impulse function, find the Laplace transform of $f(t) = t - a$

```
>> syms t s
```

```
>> syms a positive
```

```
>> laplace(dirac(t-a), t, s)
```

Output: $\exp(-a*s)$

Dirac delta Functions

Using unit impulse function, find the Laplace transform of $f(t) = (t - 1)$

```
>> syms t s
```

```
>> syms a positive
```

```
>> laplace(dirac(t-1),t,s)
```

Output: $\exp(-s)$

Dirac delta Functions

Using unit impulse function, find the Laplace transform of $f(t) = \left(t - \frac{\pi}{4}\right)$

```
>> syms t s
```

```
>> syms a positive
```

```
>> laplace(dirac(t-pi/4),t,s)
```

Output: $\exp(-(\pi*s)/4)$

Plot a 3D graph of a given Laplace Transform of a function:

- Let's say I have a function called $f(t)$ in time domain as:

$$f(t) = e^{-3t} \cos(5t)$$

- The Laplace transform of $f(t)$ becomes: $F(s) = s + 3 / (s + 3)^2 + 25$
- We want to plot of $F(s)$ as a surface above the s -plane.

```
>> [x,y]=meshgrid(-10:0.1:10);
```

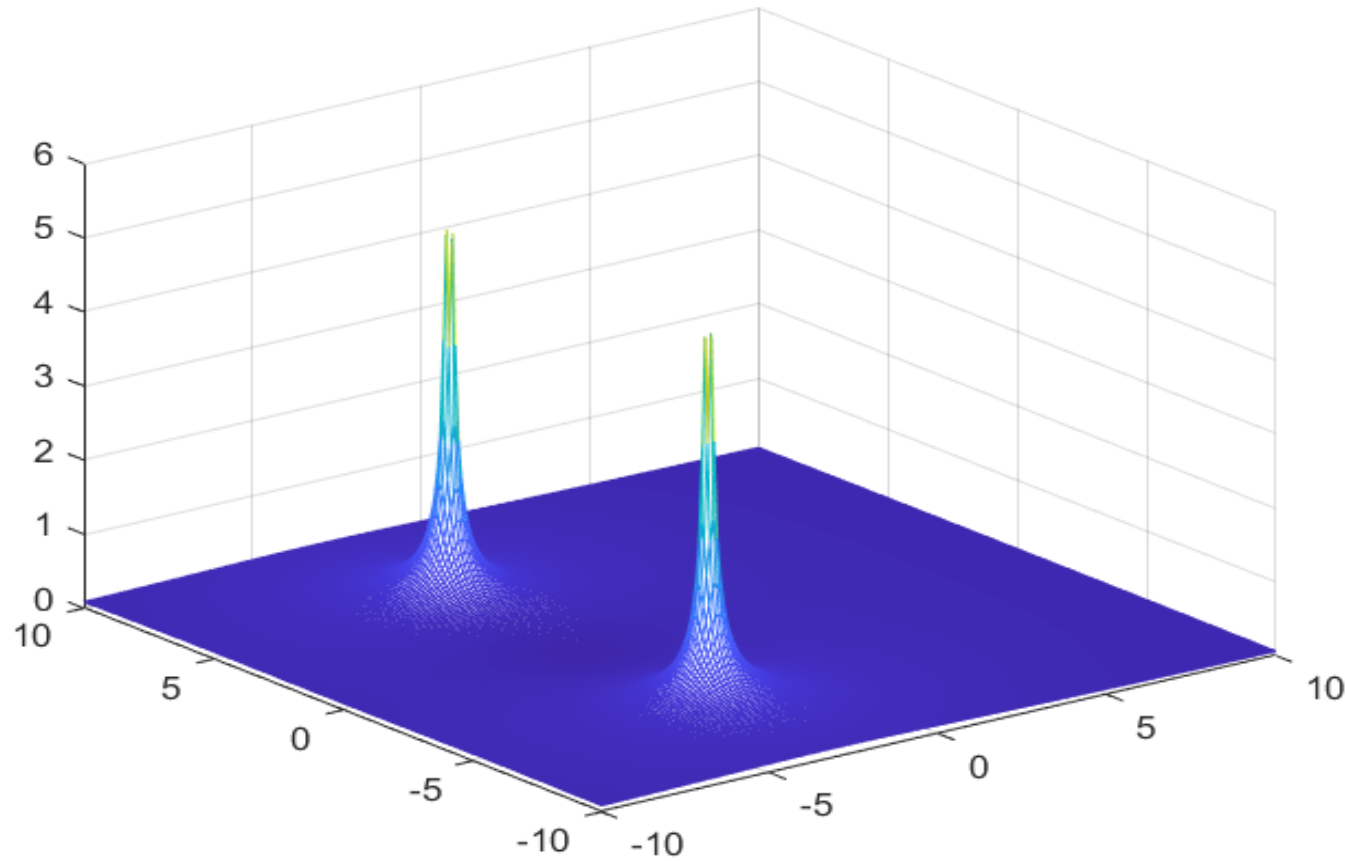
```
>> s=x+i*y;
```

```
>> z= abs((s+3)./((s+3).^2+25));
```

```
>> mesh(x,y,z)
```

Plot a 3D graph of a given Laplace Transform of a function continu:

Output:



Inverse Laplace Transform

The command for ILT `ilaplace`.

Find the inverse Laplace transforms of $F(s) = \frac{s-5}{s(s+2)^2}$

```
>> syms t s
```

```
>> F= (s-5) / (s*(s+2)^2);
```

```
>> ilaplace (F)
```

Output: $(5*\exp(-2*t))/4 + (7*t*\exp(-2*t))/2 - 5/4$

```
>> simplify(ans)
```

Output: $-5/4 + 7/2 * t * \exp(-2*t) + 5/4 * \exp(-2*t)$

Inverse Laplace Transform

Find the inverse Laplace transforms of $F(s) = \frac{10(s+2)}{s(s^2+4s+5)}$

```
>> syms t s
```

```
>> F = 10*(s+2)/(s*(s^2+4*s+5));
```

```
>> ilaplace(F)
```

Output: $-4*\exp(-2*t)*\cos(t)+2*\exp(-2*t)*\sin(t)+4$

Inverse Laplace Transform

Find the inverse Laplace transforms of $F(s) = \frac{4/s}{4+s^2}$

```
>> syms s t
```

```
>> f=(4/s)/(4+s^2);
```

```
>> ilaplace(f,s,t)
```

Output: 1-cos2*t

Inverse Laplace Transform

Find the inverse Laplace transforms of $F(s) = \frac{a}{(s^2 - a^2)}$

```
>> syms s a t
```

```
>> f= a/(s^2-a^2);
```

```
>> r=ilaplace(f,s,t)
```

Output: $\exp(a*t)/2 - \exp(-a*t)/2$

Inverse Laplace Transform

Find the inverse Laplace transforms of $\frac{s+2}{s^2-4s+13}$

```
>> syms s t
```

```
>> f= (s+2)/(s^2-4*s+13);
```

```
>> r=ilaplace(f,s,t)
```

Output: $r = \exp(2*t) * (\cos(3*t) + (4*\sin(3*t))/3)$

Solution of ODE's using Laplace transform

- This approach works only for linear differential equations with constant coefficients.
- Right-hand side functions which are sums and products of polynomials, exponential functions, sine and cosine functions, Heaviside(unit step) functions, Dirac (impulse) “functions”, initial conditions given at $t=0$.
- The main advantage is that we can handle right-hand side functions which are piecewise defined, and which contain Dirac impulse “functions”.

Solution of ODE's using Laplace transform

Consider the initial value problem $y'' + 3y' + 2y = e^{-t}$, $y(0) = 4$, $y'(0) = 5$

Define the necessary symbolic variables:

```
>> syms s t Y
```

Define the right-hand side function and find its Laplace transform:

```
>> f = exp(-t)
```

```
>> F = laplace(f, t, s)
```

Find the Laplace transform of $y'(t)$: $Y1 = sY - y(0)$.

```
>> Y1 = s*Y - 4
```

Solution of ODE's using Laplace transform

- Find the Laplace transform of $y''(t)$: $Y_2 = sY_1 - y'(0)$.

$$>> Y_2 = sY_1 - 5$$

- Set the Laplace transform of the left hand side minus the right hand side to zero and solve for Y :

$$>> \text{Sol} = \text{solve}(Y_2 + 3Y_1 + 2Y - F, Y)$$

$$\text{ans} = (4s + 1/(s + 1) + 17)/(s^2 + 3s + 2)$$

- Find the inverse Laplace transform of the solution:

$$\text{ilaplace}(\text{ans}, s, t)$$

$$\text{ans} = 12\exp(-t) - 8\exp(-2t) + t\exp(-t)$$



THANK YOU
